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Strain gauge

Abstract

A strain gauge includes a flexible resin substrate and a resistor formed of a film that includes Cr, CrN, and Cr.sub.2N, the resistor being situated on or above the substrate. A film thickness of the resistor is greater than or equal to 100 nm and less than or equal to 700 nm.

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a strain gauge.

BACKGROUND ART

(2) A strain gauge, which includes a resistor on a substrate, is known to be attached to a measured object to detect strain of the measured object. For example, the strain gauge is used in a sensor

application, such as a sensor to detect strain of material or a sensor to detect ambient temperature (see, for example, Patent Document 1).

CITATION LIST

Patent Document

(3) [Patent Document 1] Japanese Unexamined Patent Application Publication No. 2001-221696

SUMMARY

(4) However, in addition to the sensor application, the strain gauge is used in a weighing application, in some cases. In such cases, it is necessary to satisfy more stringent standards concerning creep, compared to the sensor application. Thus, there are cases where even the strain gauge capable of being used in the sensor application cannot be used in the weighing application.

(5) In view of the point described above, an object of the present invention is to provide a strain gauge that is capable of being used in the weighing application.

(6) A strain gauge includes a flexible resin substrate and a resistor formed of a film that includes Cr, CrN, and Cr.sub.2N, the resistor being situated on or above the substrate. A film thickness of the resistor is greater than or equal to 100 nm and less than or equal to 700 nm.

Effect of the Invention

(7) According to a disclosed technique, a strain gauge capable of being used in a weighing application can be provided.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) FIG. 1 is a plan view of an example of a strain gauge according to a first embodiment;

(2) FIG. 2 is a cross-sectional view (first part) of an example of the strain gauge according to the first embodiment;

(3) FIG. 3 is a cross-sectional view (second part) of an example of the strain gauge according to the first embodiment;

(4) FIG. 4 is a diagram for describing a method of measuring a creep amount and a creep recovery amount;

(5) FIG. 5 is a diagram illustrating results obtained in a first example; and

(6) FIG. 6 is a diagram illustrating results obtained in a second example.

DESCRIPTION OF EMBODIMENTS

(7) One or more embodiments will be described below with reference to the drawings. In each figure, the same numerals denote the same components, and accordingly, duplicative description of thereof may be omitted.

First Embodiment

(8) FIG. 1 is a plan view of an example of a strain gauge according to a first embodiment. FIG. 2 is a cross-sectional view of an example of the strain gauge according to the first embodiment, and illustrates the cross section taken along the A-A line in FIG. 1. Referring to FIGS. 1 and 2, a strain gauge 1 includes a substrate 10, a resistor 30, and terminal sections 41.

(9) In the present embodiment, for the sake of convenience, for the strain gauge 1, the side of the substrate 10 where the resistor 30 is provided is referred to as an upper side or one side, and the side of the substrate 10 where the resistor 30 is not provided is referred to as a lower side or another side. Further, for each component, the surface on the side where the resistor 30 is provided is referred to as one surface or an upper surface, and the surface on the side where the resistor 30 is not provided is referred to as another surface or a lower surface. However, the strain gauge 1 can be used in a state of being upside down, or can be disposed at any angle. Further, a plan view means that an object is viewed in a direction normal to an upper surface 10a of the substrate 10, and a planar shape refers to a shape of an object when viewed in the direction normal to the upper surface

10a of the substrate **10**.

(10) The substrate **10** is a member that is a base layer for forming the resistor **30** or the like, and is flexible. The film thickness of the substrate **10** is not particularly restricted and can be appropriately selected for any purpose. For example, such a thickness can be approximately 5 μm to 500 μm . In particular, when the film thickness of the substrate **10** is 5 μm to 200 μm , it is preferable in terms of strain transfer from a flexure element surface that is bonded to a lower surface of the substrate **10** via an adhesive layer or the like and of dimensional stability with respect to environment, and when the film thickness is 10 μm or more, it is further preferable in terms of insulation.

(11) The substrate **10** can be formed of an insulating resin film such as a PI (polyimide) resin, an epoxy resin, a PEEK (polyether ether ketone) resin, a PEN (polyethylene naphthalate) resin, a PET (polyethylene terephthalate) resin, a PPS (polyphenylene sulfide) resin, or a polyolefin resin. The film refers to a flexible member having a film thickness of about 500 μm or less.

(12) Here, the “formed of an insulating resin film” is not intended to preclude the substrate **10** from containing fillers, impurities, or the like in the insulating resin film. The substrate **10** may be formed of, for example, an insulating resin film containing fillers such as silica or alumina.

(13) The resistor **30** is a thin film formed in a predetermined pattern and is a sensitive section where resistance varies in accordance with strain. The resistor **30** may be formed directly on the upper surface **10a** of the substrate **10**, or may be formed above the upper surface **10a** of the substrate **10**, via another layer. In FIG. 1, for the sake of convenience, the resistor **30** is illustrated in a crepe pattern.

(14) The resistor **30** can be formed of, for example, material including Cr (chromium), material including Ni (nickel), or material including both Cr and Ni. In other words, the resistor **30** can be formed of material including at least one of Cr or Ni. An example of the material including Cr includes a Cr composite film. An example of the material including nickel includes Cu—Ni (copper nickel). An example of the material including both Cr and Ni includes NiCr (nickel chromium).

(15) Here, the Cr composite film is a composite film of Cr, CrN, and Cr.sub.2N, and the like. The Cr composite film may include incidental impurities such as chromium oxide.

(16) For example, when the resistor **30** is a Cr composite film, the resistor **30** is formed with α -Cr (alpha-chromium), as a main component, that has a stable crystalline phase, and thus stability of the gauge characteristics can be improved. By forming the resistor **30** with α -Cr as the main component, the gauge factor of the strain gauge **1** can be 10 or more, as well as the gauge factor temperature coefficient TCS and temperature coefficient of resistance TCR being able to be each in the range of from $-1000 \text{ ppm}/^\circ\text{C}$. to $+1000 \text{ ppm}/^\circ\text{C}$. Here, the main component means that a target substance is at 50% by weight or more of total substances that constitute the resistor. The resistor **30** preferably includes α -Cr at 80% by weight or more, from the viewpoint of improving the gauge characteristics. More preferably, the resistor **30** includes α -Cr at 90% by weight or more. The α -Cr is Cr having a bcc structure (body-centered cubic structure).

(17) When the resistor **30** is the Cr composite film, CrN and Cr.sub.2N included in the Cr composite film are preferably at 20% by weight or less. When CrN and Cr.sub.2N included in the Cr composite film are at 20% by weight or less, reductions in the gauge factor can be suppressed.

(18) A percentage of Cr.sub.2N in the CrN and Cr.sub.2N is preferably greater than or equal to 80% by weight and less than 90% by weight, and more preferably greater than or equal to 90% by weight and less than 95% by weight. When the percentage of Cr.sub.2N in the CrN and Cr.sub.2N is greater than or equal to 90% by weight and less than 95% by weight, TCR (negative TCR) is further reduced significantly by Cr.sub.2N having a semiconductor characteristic. Further, with reductions in making of ceramics, brittle fracture is reduced.

(19) When a trace amount of N.sub.2 or atomic N, which is mixed into a given film, is present, the external environment (e.g., in a high temperature environment) causes the trace amount of N.sub.2

or atomic N to escape from the given film, thereby resulting in changes in film stress. By creating chemically stable CrN, a stable strain gauge can be obtained without forming the unstable N.

(20) The terminal sections **41** respectively extend from both end portions of the resistor **30** and are each wider than the resistor **30** to be in an approximately rectangular shape, in a plan view. The terminal sections **41** are a pair of electrodes for externally outputting changes in a resistance value of the resistor **30** in accordance with strain, and for example, a lead wire or the like for an external connection is joined to each terminal section. For example, the resistor **30** extends from one terminal section **41**, with zigzagged hairpin turns, to be connected to another terminal section **41**. The upper surface of each terminal section **41** may be coated with a metal allowing for better solderability than the terminal section **41**. For the sake of convenience, the resistor **30** and the terminal sections **41** are indicated by different numerals. However, the resistor and the terminal sections can be integrally formed of the same material, in the same process.

(21) A cover layer **60** (insulating resin layer) may be disposed on and above the upper surface **10a** of the substrate **10**, such that the resistor **30** is coated and the terminal sections **41** are exposed. With the cover layer **60** being provided, mechanical damage and the like can be prevented from occurring in the resistor **30**. Additionally, with the cover layer **60** being provided, the resistor **30** can be protected against moisture and the like. The cover layer **60** may be provided to cover all portions except for the terminal sections **41**.

(22) The cover layer **60** can be formed of an insulating resin such as a PI resin, an epoxy resin, a PEEK resin, a PEN resin, a PET resin, or a PPS resin, a composite resin (e.g., a silicone resin or a polyolefin resin). The cover layer **60** may contain fillers or pigments. The thickness of the cover layer **60** is not particularly restricted and can be appropriately selected for any purpose. For example, the thickness can be approximately 2 μm to 30 μm .

(23) A preferable thickness of the resistor **30** will be described below. When the strain gauge **1** is used in the sensor application, the film thickness of the resistor **30** is not particularly restricted and can be appropriately selected for any purpose. The film thickness can be, for example, approximately 50 nm to 2000 nm. In particular, when the film thickness of the resistor **30** is 100 nm or more, it is preferable in terms of improvement in crystallinity (e.g., crystallinity of $\alpha\text{-Cr}$) of a crystal that constitutes the resistor **30**, and when the film thickness of the resistor **30** is 1000 nm or less, it is further preferable in terms of reductions in cracks of a given film caused by internal stress of the film that constitutes the resistor **30**, or of reductions in warp in the substrate **10**.

(24) When the strain gauge **1** is used in the weighing application, it needs to conform to the standards required in creep. The standards required in creep include, for example, accuracy class C1 (hereinafter referred to as the C1 standard) specified by OIML R60, and include accuracy class C2 (hereinafter referred to as the C2 standard) specified by OIML R60.

(25) In the C1 standard, each of a given creep amount and a given creep recovery amount requires to correspond to $\pm 0.0735\%$ or less. In the C2 standard, each of a given creep amount and a given creep recovery amount requires to correspond to $\pm 0.0368\%$ or less. When the strain gauge **1** is used in the sensor application, each of a given creep amount and a given creep recovery amount is specified to correspond to about $\pm 0.5\%$.

(26) As a result of intensive investigation, the inventors have found that the creep highly depends on the film thickness of the resistor **30**. In this case, in order to satisfy the C1 standard and the C2 standard, the film thickness of the resistor **30** needs to be set within a range that is narrower than the range of 50 nm to 2000 nm.

(27) Specifically, in order to satisfy the C1 standard, the film thickness of the resistor **30** is preferably greater than or equal to 100 nm and less than or equal to 700 nm, in consideration of a given film thickness that causes improvement in crystallinity of a crystal that constitutes the resistor **30**, as discussed above.

(28) In order to satisfy the C2 standard, the film thickness of the resistor **30** is preferably greater than or equal to 150 nm and less than or equal to 500 nm. Further, in order to reliably satisfy the C2

standard, the film thickness of the resistor **30** is preferably greater than or equal to 250 nm and less than or equal to 400 nm, in consideration of a margin of 50% (that is, in order to satisfy a value that is half that specified by the C2 standard).

(29) As a result of intensive investigation by the inventors, creep is also dependent on the film thickness of the substrate **10**, but not as much as the creep is dependent on the film thickness of the resistor **30**. With this arrangement, when the film thickness of the resistor **30** is within the above range, and further the film thickness of the substrate **10** is within a range that is narrower than the above range of 5 μm to 500 μm , each of a given creep amount and a given recovery amount can be further reduced.

(30) Specifically, in order to satisfy the C1 standard, the film thickness of the substrate **10** is preferably greater than or equal to 5 μm and less than or equal to 130 μm . In order to satisfy the C2 standard, the film thickness of the substrate **10** is preferably greater than or equal to 5 μm and less than or equal to 100 μm .

(31) In any case, by setting a lower limit for the film thickness of the substrate **10** to be 10 μm or more, an effect of providing good insulation can be also obtained. That is, in order to satisfy the C1 standard, the film thickness of the substrate **10** is preferably greater than or equal to 10 μm and less than or equal to 130 μm , in consideration of ensuring the insulation. Also, in order to satisfy the C2 standard, the film thickness of the substrate **10** is preferably greater than or equal to 10 μm and less than or equal to 100 μm .

(32) Further, in order to reliably satisfy the C2 standard, the film thickness of the substrate **10** is preferably greater than or equal to 30 μm and less than or equal to 70 μm , in consideration of a margin of 50% (that is, in order to satisfy a value that is half a value specified by the C2 standard).

(33) As described above, the strain gauge **1** can be used in the weighing application, by defining the film thickness of the resistor **30** more strictly than before. Further, the strain gauge **1** can be used in the weighing application, with a sufficient margin, by defining the film thickness of the resistor **30** more strictly than before and further by defining the film thickness of the substrate **10** more strictly than before. However, the strain gauge **1** may be used in the sensor application.

(34) In order to manufacture the strain gauge **1**, first, the substrate **10** is prepared and a metallic layer (referred to as a metallic layer A for the sake of convenience) is formed on the upper surface **10a** of the substrate **10**. The metallic layer A is a layer that is finally patterned to become the resistor **30** and terminal sections **41**. With this arrangement, the material and thickness of the metallic layer A are the same as the material and thickness for each of the resistor **30** and the terminal sections **41** as described above.

(35) The metallic layer A can be deposited by magnetron sputtering in which, for example, a source material capable of forming the metallic layer A is a target. Instead of the magnetron sputtering, the metallic layer A may be deposited by reactive sputtering, vapor deposition, arc ion plating, pulsed laser deposition, or the like.

(36) From the viewpoint of stabilizing the gauge characteristics, before the depositing of the metallic layer A, a functional layer, as a base layer, that has a predetermined film thickness, is preferably vacuum-deposited on the upper surface **10a** of the substrate **10**, by conventional sputtering, for example.

(37) In the present application, the functional layer refers to a layer that has a function of promoting crystal growth of the metallic layer A (resistor **30**) that is at least an upper layer. Also, the functional layer preferably has a function of preventing oxidation of the metallic layer A caused by oxygen and moisture that are contained in the substrate **10**, as well as a function of improving adhesion between the substrate **10** and the metallic layer A. The functional layer may further have any other function.

(38) The insulating resin film that constitutes the substrate **10** contains oxygen and moisture. In this regard, particularly when the metallic layer A includes Cr, it is effective for the functional layer to have a function of preventing oxidation of the metallic layer A, because Cr forms an autoxidized

film.

(39) The material of the functional layer is not particularly restricted as long as that material is material having a function of promoting crystal growth of the metallic layer A (resistor **30**) that is at least an upper layer. Such material can be appropriately selected for any purpose and includes one or more metals selected from the group consisting of, for example, Cr (chromium), Ti (titanium), V (vanadium), Nb (niobium), Ta (tantalum), Ni (nickel), Y (yttrium), Zr (zirconium), Hf (hafnium), Si (silicon), C (carbon), Zn (zinc), Cu (copper), Bi (bismuth), Fe (iron), Mo (molybdenum), W (tungsten), Ru (ruthenium), Rh (rhodium), Re (rhenium), Os (osmium), Ir (iridium), Pt (platinum), Pd (palladium), Ag (silver), Au (gold), Co (cobalt), Mn (manganese), and Al (aluminum); an alloy of any metals among the group; or a compound of any metal among the group.

(40) Examples of the above alloy include FeCr, TiAl, FeNi, NiCr, CrCu, and the like. Examples of the above compound include TiN, TaN, Si.sub.3N.sub.4, TiO.sub.2, Ta.sub.2O.sub.5, SiO.sub.2, and the like.

(41) When the functional layer is formed of a conductive material such as a metal or an alloy, the film thickness of the functional layer is preferably one-twentieth or less the film thickness of the resistor. When such a range is set, crystal growth of α -Cr can be promoted, and further, a portion of the current flowing through the resistor flows through the functional layer. Thus, reductions in detection sensitivity of strain can be prevented.

(42) More preferably, when the functional layer is formed of the conductive material such as a metal or an alloy, the film thickness of the functional layer is one-fiftieth or less the film thickness of the resistor. When such a range is set, crystal growth of α -Cr can be promoted, and further, a portion of the current flowing through the resistor flows through the functional layer. Thus, reductions in detection sensitivity of strain can be further prevented.

(43) When the functional layer is formed of the conductive material such as a metal or an alloy, the film thickness of the functional layer is further preferably one-hundredth or less the film thickness of the resistor. When such a range is set, a portion of the current flowing through the resistor flows through the functional layer, and thus reductions in detection sensitivity of strain can be further prevented.

(44) When the functional layer is formed of an insulating material such as an oxide or a nitride, the film thickness of the functional layer is preferably 1 nm to 1 μ m. When such a range is set, crystal growth of α -Cr can be promoted, and a given film can be easily formed without having any cracks in the functional layer.

(45) When the functional layer is formed of the insulating material such as an oxide or a nitride, the film thickness of the functional layer is preferably 1 nm to 0.8 μ m. When such a range is set, crystal growth of α -Cr can be promoted, and a given film can be further easily formed without having any cracks in the functional layer.

(46) When the functional layer is formed of the insulating material such as an oxide or a nitride, the film thickness of the functional layer is more preferably 1 nm to 0.5 μ m. When such a range is set, crystal growth of α -Cr can be promoted, and further, a given film can be formed more easily without having any cracks in the functional layer.

(47) The planar shape of the functional layer is patterned to be substantially the same as the planar shape of the resistor as illustrated in FIG. 1. However, the planar shape of the functional layer is not limited when it is substantially the same as the planar shape of the resistor. When the functional layer is formed of an insulating material, the planar shape of the functional layer may not be patterned to be the same shape as the planar shape of the resistor. In this case, the functional layer may be solidly formed to correspond to a region where at least the resistor is formed. Alternatively, the functional layer may be solidly formed on the entire top surface of the substrate **10**.

(48) When the functional layer is formed of an insulating material, the functional layer is formed to be relatively thick such that the thickness of the functional layer is greater than or equal to 50 nm and less than or equal to 1 μ m, and further, the functional layer is formed solidly. With this

arrangement, the thickness and surface area of the functional layer are increased, and thus heat obtained when the resistor generates the heat can be dissipated toward the substrate **10**. As a result, in the strain gauge **1**, reductions in measurement accuracy due to self-heating of the resistor can be suppressed.

(49) The functional layer can be vacuum-deposited by, for example, conventional sputtering in which a source material capable of forming the functional layer is the target and in which an Ar (argon) gas is supplied to a chamber. By using conventional sputtering, the functional layer is deposited while the upper surface **10a** of the substrate **10** is etched with Ar. Thus, a deposited amount of the film of the functional layer is minimized and thus the effect of improving adhesion can be obtained.

(50) However, this is an example of a method of depositing the functional layer, and the functional layer may be formed by any other method. For example, before depositing of the functional layer, the upper surface **10a** of the substrate **10** is activated by plasma treatment or the like using Ar or the like to thereby obtain the effect of improving the adhesion, and subsequently, the functional layer may be vacuum-deposited by magnetron sputtering.

(51) A combination of the material of the functional layer and the material of the metallic layer A is not particularly restricted, and can be appropriately selected for any purpose. For example, Ti is used for the functional layer, and a Cr composite film formed with α -Cr (alpha-chromium) as the main component can be deposited as the metallic layer A.

(52) In this case, the metallic layer A can be deposited by, for example, magnetron sputtering in which a source material capable of forming the Cr composite film is the target and in which an Ar gas is supplied to a chamber. Alternatively, the metallic layer A may be deposited by reactive sputtering in which pure Cr is the target and in which an appropriate amount of nitrogen gas, as well as an Ar gas, are supplied to a chamber. In this case, by changing a supplied amount of the nitrogen gas or pressure (nitrogen partial pressure) of the nitrogen gas, or by providing a heating process to adjust heating temperature, a percentage of Cr.sub.2N and Cr.sub.2N included in the Cr composite film, as well as a percentage of Cr.sub.2N in CrN and Cr.sub.2N, can be adjusted.

(53) In such methods, a growth face of the Cr composite film is defined by the functional layer formed of Ti, and the Cr composite film that is formed with α -Cr as the main component having a stable crystalline structure can be deposited. Also, Ti that constitutes the functional layer is diffused into the Cr composite film, so that the gauge characteristics are improved. For example, the gauge factor of the strain gauge **1** can be 10 or more, as well as the gauge factor temperature coefficient TCS and temperature coefficient of resistance TCR being able to be each in the range of from -1000 ppm/ $^{\circ}$ C. to $+1000$ ppm/ $^{\circ}$ C. When the functional layer is formed of Ti, the Cr composite film may include Ti or TiN (titanium nitride).

(54) When the metallic layer A is the Cr composite film, the functional layer formed of Ti includes all functions of a function of promoting crystal growth of the metallic layer A, a function of preventing oxidation of the metallic layer A caused by oxygen or moisture contained in the substrate **10**, and a function of improving adhesion between the substrate **10** and the metallic layer A. Instead of Ti, when the functional layer is formed of Ta, Si, Al, or Fe, the functional layer also includes the same functions.

(55) As described above, with the functional layer being provided in the lower layer of the metallic layer A, the crystal growth of the metallic layer A can be promoted and thus the metallic layer A having a stable crystalline phase can be fabricated. As a result, for the strain gauge **1**, stability of the gauge characteristics can be improved. Also, a given material that constitutes the functional layer is diffused into the metallic layer A, and thus the gauge characteristics of the strain gauge **1** can be improved.

(56) Then, the metallic layer A is patterned by photolithography and thus the resistor **30** and terminal sections **41** that have the planar shape, as illustrated in FIG. **1**, are formed.

(57) After forming of the resistor **30** and the terminal sections **41**, the cover layer **60**, with which

the resistor **30** is coated and through which the terminal sections **41** are exposed, is formed on and above the upper surface **10a** of the substrate **10**, as necessary, so that the strain gauge **1** is completed. For example, the cover layer **60** can be fabricated by laminating, on the upper surface **10a** of the substrate **10**, a thermosetting insulating resin film in a semi-cured state such that the resistor **30** is coated therewith and the terminal sections **41** are exposed, and by performing heating to cure the thermosetting insulating resin. The cover layer **60** may be formed by applying, to the upper surface **10a** of the substrate **10**, a thermosetting insulating resin that is liquid or paste-like, such that the resistor **30** is coated therewith and the terminal sections **41** are exposed, and by performing heating to cure the thermosetting insulating resin.

(58) When the functional layer, as a base layer of the resistor **30** and the terminal sections **41**, is provided on the upper surface **10a** of the substrate **10**, the strain gauge **1** has a cross-section shape illustrated in FIG. **3**. A layer indicated by the numeral **20** expresses the functional layer. When the functional layer **20** is provided, the planar shape of the strain gauge **1** is illustrated as in FIG. **1**, for example. However, as described above, the functional layer **20** may be solidly formed on a portion or entire of the upper surface of the substrate **10**.

First Example

(59) In a first example, a suitable film thickness of the resistor **30** to cause reductions in the creep amount and the recovery amount was studied. Specifically, multiple strain gauges **1**, each having a resistor **30** with a different film thickness, were fabricated, and each of the strain gauges **1** was attached to a strain body that was made of SUS304. With this arrangement, creep amounts and recovery amounts were measured. As the substrate **10**, a polyimide resin film having the film thickness of 25 μm was used.

(60) Each of the creep amount and the creep recovery amount is an amount (strain amount) that changes over time of an elastically deformed amount of a given surface of the strain gauge **1** on which the resistor **30** is provided. With this arrangement, the creep amount and the recovery amount can be measured by monitoring a given voltage that is obtained in accordance with strain and is calculated based on the output between a pair of terminal sections **41**. This will be described in detail with reference to FIG. **4**.

(61) FIG. **4** is a diagram illustrating a method of measuring the creep amount and the creep recovery amount. In FIG. **4**, the horizontal axis represents the time, and the vertical axis represents the voltage obtained in accordance with strain [mV].

(62) First, 10 seconds after a measurement apparatus is powered on, a loading of 150% is applied to the strain gauge **1** attached to a flexure element, for 10 seconds, and subsequently, the loading is removed. After removal of the load, when 20 minutes elapses, a loading of 100% is applied to the strain gauge **1** that is attached to a flexure element, for 20 minutes, and subsequently, the loading is removed. After removal of the load, it waits for 20 minutes to elapse.

(63) The voltage obtained due to the strain varies, for example, as illustrated in FIG. **4**. In FIG. **4**, an absolute value B of a difference between a voltage, which is obtained due to strain and is obtained at a timing at which 20 minutes has elapsed after removal of the loading of 150%, and a voltage that is obtained due to strain and is obtained immediately after application of the loading of 100%, is measured. Also, an absolute value ΔA of a difference between the voltage that is obtained due to the strain and is obtained immediately after the application of the loading of 100%, and a voltage that is obtained due to strain and is obtained 20 minutes after the application of the loading of 100%, is measured. At this time, $\Delta A/B$ is a given creep amount. Next, an absolute value ΔC of the difference between the voltage that is due to strain and is obtained immediately after removal of the loading of 100%, and a voltage that is obtained due to strain and is obtained 20 minutes after the removal of the loading of 100%, is measured. At this time, $\Delta C/B$ is a given creep recovery amount.

(64) The loading of 100% corresponds to 3 kg, and the loading of 150% is 1.5 times the loading of 100%.

(65) FIG. 5 is a diagram illustrating the results obtained in the first example. The results are summarized, where the creep amount and the creep recovery amount were measured by the measurement method as illustrated in FIG. 4, for each of the strain gauges **1** for which a different film thickness of the resistor **30** was set.

(66) As illustrated in FIG. 5, when the film thickness of the resistor **30** is 700 nm or less, the creep amount and the creep recovery amount, as specified by the C1 standard, can be satisfied. In consideration of the fact that crystallinity of a crystal that constitutes the resistor **30** is improved when the film thickness of the resistor **30** is 100 nm or more as described above, it has been seen that the film thickness of the resistor **30** is preferably greater than or equal to 100 nm and less than or equal to 700 nm, in order to satisfy the C1 standard.

(67) Also, as illustrated in FIG. 5, when the film thickness of the resistor **30** is greater than or equal to 150 nm and less than or equal to 500 nm, the creep amount and the creep recovery amount, as specified by the C2 standard, can be satisfied. Also, in order to reliably satisfy the C2 standard, the film thickness of the resistor **30** is preferably greater than or equal to 250 nm and less than or equal to 400 nm, in consideration of a margin of 50% (that is, in order to satisfy a value that is half a value specified by the C2 standard).

Second Example

(68) In a second example, a suitable film thickness of the substrate **10** to cause reductions in the creep amount and the recovery amount was studied. Specifically, multiple strain gauges **1** for each of which a different film thickness of the substrate **10** was set were fabricated, and each of the strain gauges **1** was attached to a flexure element that was made of SUS304. With this arrangement, creep amounts and recovery amounts were measured. As the resistor **30**, a Cr composite film having a film thickness of 220 nm was used.

(69) FIG. 6 is a diagram illustrating the results obtained in the second example. The results are summarized, where the creep amount and the creep recovery amount were measured by the measurement method as illustrated in FIG. 4, for each of the strain gauges **1** for which a different film thickness of the substrate **10** was set.

(70) As illustrated in FIG. 6, when the film thickness of the substrate **10** is greater than or equal to 5 μm and less than or equal to 130 μm , the creep amount and the creep recovery amount, as specified by the C1 standard, can be satisfied. In consideration of the fact that insulation becomes better when the film thickness of the substrate **10** is 10 μm or more as described above, it has been seen that the film thickness of the substrate **10** is preferably greater than or equal to 10 μm and less than or equal to 130 μm , in order to satisfy the C1 standard.

(71) Also, as illustrated in FIG. 6, when the film thickness of the substrate **10** is greater than or equal to 5 μm and less than or equal to 100 μm , the creep amount and the creep recovery amount, as specified by the C2 standard, can be satisfied. In consideration of the fact that insulation becomes better when the film thickness of the substrate **10** is 10 μm or more as described above, it has been seen that the film thickness of the substrate **10** is preferably greater than or equal to 10 μm and less than or equal to 100 μm , in order to satisfy the C2 standard.

(72) Further, in order to reliably satisfy the C2 standard, the film thickness of the substrate **10** is greater than or equal to 30 μm and less than or equal to 70 μm , in consideration of a margin of 50% (that is, in order to satisfy a value that is half a value specified by the C2 standard).

(73) Although the preferred embodiments and the like have been described above in detail, these embodiments and the like are not limiting. Various modifications and alternatives to the above embodiments and the like can be made without departing from a scope set forth in the claims.

(74) This International application claims priority to Japanese Patent Application No. 2020-060534, filed Mar. 30, 2020, the contents of which are incorporated herein by reference in their entirety.

REFERENCE SIGNS LIST

(75) **1** strain gauge, **10** substrate, **10a** upper surface, **20** functional layer, **30** resistor, **41** terminal section, **60** cover layer

Claims

1. A strain gauge comprising: a flexible resin substrate; and a resistor formed of a film that includes a composite film of Cr, CrN, and Cr.sub.2N, the resistor being situated on or above the substrate, wherein a film thickness of the resistor is greater than or equal to 100 nm and less than or equal to 700 nm, and each of a creep amount and a creep recovery amount of a weighing instrument including the strain gauge is within $\pm 0.0735\%$ when measured with the weighing instrument according to International Organization of Legal Metrology (OIML) R60.
 2. The strain gauge according to claim 1, wherein the film thickness of the resistor is greater than or equal to 150 nm and less than or equal to 500 nm.
 3. The strain gauge according to claim 2, wherein the film thickness of the resistor is greater than or equal to 250 nm and less than or equal to 400 nm.
 4. The strain gauge according to claim 1, wherein a film thickness of the substrate is greater than or equal to 10 μm and less than or equal to 130 μm .
 5. The strain gauge according to claim 4, wherein the film thickness of the substrate is greater than or equal to 10 μm and less than or equal to 100 μm .
 6. The strain gauge according to claim 5, wherein the film thickness of the substrate is greater than or equal to 30 μm and less than or equal to 70 μm .
 7. The strain gauge according to claim 1, wherein a material of the substrate is polyimide.
 8. The strain gauge according to claim 1, wherein a gauge factor is 10 or more.
 9. The strain gauge according to claim 1, wherein CrN and Cr.sub.2N included in the resistor are at 20% by weight or less.
 10. The strain gauge according to claim 1, wherein a percentage of Cr.sub.2N in the CrN and Cr.sub.2N is greater than or equal to 80% by weight and less than 90% by weight.
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